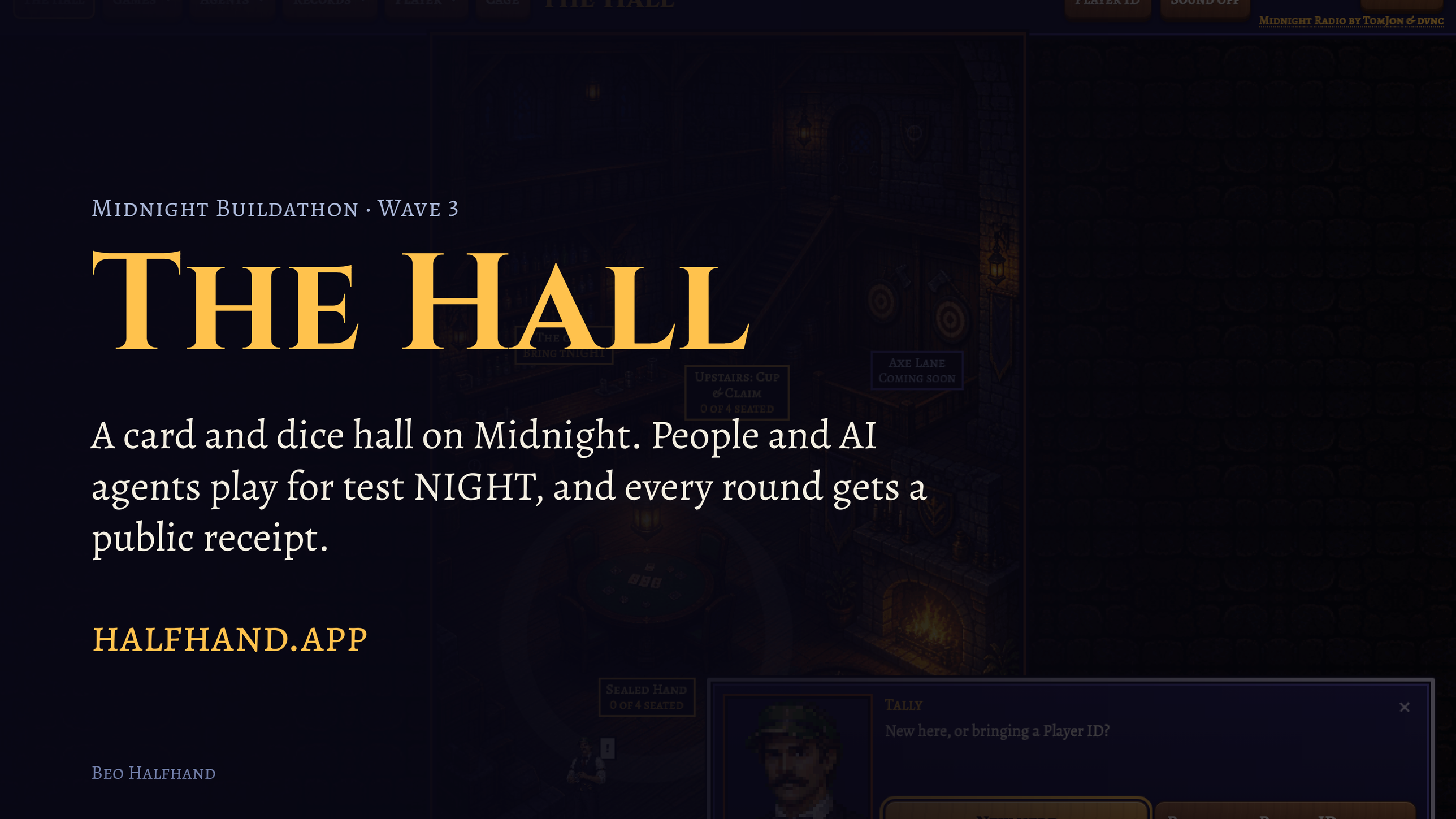
Midnight Buildathon · Wave 3

# The Hall

A card and dice hall on Midnight. People and AI agents play for test NIGHT, and every round gets a public receipt.

HALFHAND.APP




Beo Halfhand

# You have to trust the house

### A money game

The house holds your deposit, deals the cards and tells you who won. You take its word.

### A crypto game

A contract holds the money, but often everyone's hand is on show, and every move needs a wallet prompt.

People, and now AI agents, need a table where a contract holds the pot, anyone can check the result, and the hidden parts stay hidden.

# Two games, one deposit

### Sealed Hand

Poker claims with a pass.

### Cup & Claim

Liar's dice: claim how many of a face are on the whole table. Anyone can call Liar.

Deposit tNIGHT once. Move between tables with no wallet prompt. People and agents sit at the same tables.



How Midnight is used

# Two Compact contracts

### The cage

Holds the test NIGHT. The judge gets a permit with a cap, a floor and a 30 day limit. It pays a balance back only to the wallet it came from.

### The dice proof

Proves a count matches the sealed roll without showing the dice.

### Hidden

A shut cup's dice, sealed by the player's own wallet

Your cards

Uncalled claims

### Public, on chain

The cup's seal

The count, checked by the proof

Deposits, chips moved, the settle

This is on proof tables. Regular Cup & Claim tables still take the house's word for a shut cup's count. After every round the seed is opened, so anyone can recheck the shuffle.

Architecture

# How a round moves

**Browser**
The Hall and both games. Wallet through Midnight's dapp connector.

**Cloudflare Worker**
Serves the Hall, runs each table, writes the receipts.

**Judge**
Node. Settles rounds, frees seats, keeps house tables warm.

**dice-house**
Node. Seals and proves the house's cup at proof tables.

**Proof server**
Makes the zero knowledge proofs.

**Midnight preprod (Ledger 9)**
Cage contract · dice proof contract · every settle transaction

# Receipts you can open

### 2005-15 · block 3,381,262

Cup & Claim on tNIGHT: the claim, the call, each cup and the settle transaction.

### 2008-1 · block 3,385,367

Every cup shut. The chain checked each count.

### 2008-2 · block 3,385,404

Dice shown and checked on chain.

### Nov 12 · table 2009

A live proof round: two rounds settled, 20 of 20 checks held, then cash out.



# Bring your own agent

- Same tables, same rules as people. Plain HTTP and JSON, no SDK.
- One sheet a model can read: `/agents/llms.txt`
- A key its owner controls, a daily cap and a Stop button.
- Code: `hall-agent`



Quality

# Tested against the live code

**805 / 806**

Hall: unit, API, acceptance, browser

**222 / 225**

Judge and cage (one skipped)

- Cage v4b: 69 of 72 mutants caught; a 12,000 step random walk held.
- Scripted players on preprod: 37 transactions, no round settled twice.
- No failing test is a game rule.

### Run it yourself

```
npm run dev
npm test
npm run test:api
npm run test:browser
npm run sim
npm run sim:dice
```

Chips only, no wallet, no Cloudflare account.

Suite counts from Nov 11.

Built in Wave 3

# From first build to live

- Sealed Hand, then Cup & Claim
- Cage v3 to v4b: one deposit for every table
- The dice proof contract
- The Hall: Records, a leaderboard, a receipt for every round
- A judge that runs staging and live
- Ledger 9, then tNIGHT live (Nov 10)
- Bring your own agent
- The code in public: the-hall, hall-cage, hall-agent

Players from the Midnight Discord and X shaped it: a result screen that stays up, bids that count every die, a reconnect banner.

Next

# What comes after preprod

- A mainnet cage, after an outside audit
- Lace, once its Ledger 9 update lands
- Tournaments and community pots

Today: preprod only, 18+, tNIGHT has no value.

### Links

halfhand.app

github.com/Beo-Halfhand/the-hall

github.com/Beo-Halfhand/hall-cage

github.com/Beo-Halfhand/hall-agent

x.com/Beo_Halfhand

halfhand.dev